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Description - This procedure explains the functional checks of the out put power from the RF Amplifiers for the 1.0T RF/PDU cabinet.

CAUTION

Possible equipment damage. Do not operate the RF amplifier with an improper load connected to its output. Doing so may permanently damage the amplifier.

1- OVERVIEW

This product is intended specifically for 1.0T medical MRI, and has the following characteristics.

A) Working frequency of 42.68MHz ± 225KHz
B) Pulse output only--Can not be used under continuous wave
C) Peak output of 4kw, Unblank pulse width of 30m sec. max., Duty of 60% max.
D) RF output noise level under blank condition of less than 160dBm/Hz

2- RF AMPLIFIER MAXIMUM POWER CHECKS

2-1 Tools Required

Refer to Table 2-1.

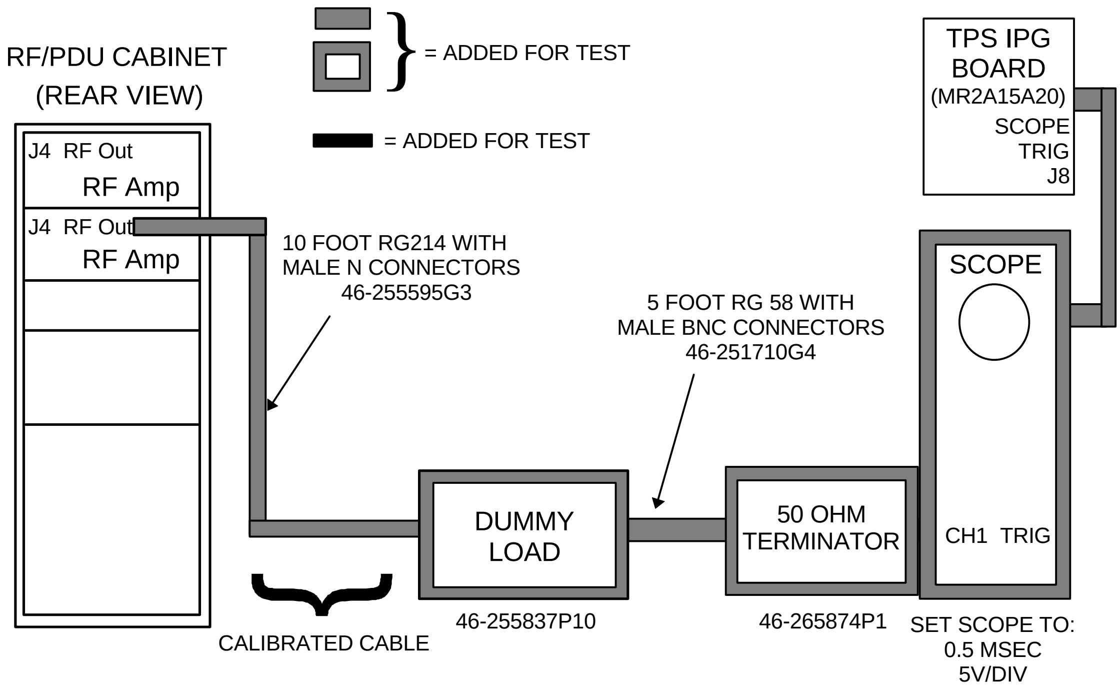
TABLE 2-1
TOOLS REQUIRED

Item	Description	Part Number	Qty.
1.	RF Test Cables Kit	46-255816G1	1
2.	50-ohm dummy load, 200 watt, 30 dB attenuator Bird Model 8322	46-255837P10	1
3.	100mHz Scope - Tektronix 468 or equivalent	46-183029P61 or P64	1
4.	RF Power Measurement Kit	46-317724G1 or G2	1

2-2 (Body Max Pwr) Body Mode RF Maximum Power Output

1. Disconnect the input cable (J3, RF In) on the Amp that **is not** being checked.

2. Connect the system as shown in Illustration 2-1.



CONFIGURATION FOR POWER MEASUREMENTS USING OSCILLOSCOPE
ILLUSTRATION 2-1

Note
The Dummy Load is connected to the RF Amplifier that is being checked.

2-2- (Body Max Pwr) Body Mode RF Maximum Power Output (continued)

3. Prepare the system to scan in body mode, as shown in Table 2-2.

TABLE 2-2
SCAN PROTOCOL: BODY MODE

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

1. (If you haven't already done so)
   Click on **[New Pt]**, and enter
   Id: **geservice**
   Set Patient Protocols to **Service**.
   Name: **rfamp cal body**
   Weight (Lb.): **300**
   At front enclosure, press **LANDMARK**, then **MOVE TO SCAN**.
2. In the protocol field, type **o.41** (o=Other, 41 =series) to load the protocol.
3. **[Save Series]**.
4. Select **[Research Operations]** with the mouse cursor, and right click. Then select **[Setup Params]**. Set TG to **50** (transmit gain=90dB).

In the next steps you will change the CV Values as indicated. See the User Interface Tutorial for specific help with CVs.

5. Select **[Research Operations]** with the mouse cursor, and right click. Then select **[Display CVs]**.
6. Set value of CV calmode to **5** (Dual Logamp Waveform).
7. Select **[Accept]**.
8. Set value of CV ia_rf1 to **32766** (sets 90º pulse full scale).
9. Select **[Accept]**.
10. Set value of CV ia_rf2 to **0** (turns off 180º pulse).
11. Select **[Accept]**.
12. Select **[Research Operations]**, then **[Download]**, and **[Manual Prescan]**.

If you need help, see the User Interface Tutorial.

4. Disable TR driver faults (switch down) on front of Erbtec System Support Module (SSM).

5. Select **[Manual Prescan], [Scan TR (R1/R2)]**. While watching the RF output power on the scope, gradually increase TG to 200.

6. Peak output power of 4kw or 66dBm should be measured.

7. Repeat section *2-2 (BODY MAX PWR) BODY MODE RF MAXIMUM POWER OUTPUT,* for the other amp to measure peak output power.

If the RF Amps power measurements are in appropriate ranges, refer to the *RF Interface (RFI) Module Functional Checks* procedure.

If either one or both RF Amp(s) is not measuring 4kw after checking the RFI module and exciter, then refer to *RF Amplifier Replacement Procedures* to replace RF Amplifier(s).

3- FAN FUNCTIONAL CHECKS

1. Visually check that fans are rotating and that air is blowing out off the amplifier.

Note

If fans are not functioning properly the FAULT HEAT LED, on front panel of RF Amplifier, may be lit.

2. If air is not coming out of the back of the amplifier, check the installation of fans for proper air flow direction. See *RF Amplifier Replacement Procedures*.

3. If fans are not rotating, check the following:

   a) Check ventilation – make sure ventilation holes are not blocked.
   b) Check fuses – make sure fuses are not broken.
   c) Check AC connections – make sure fans are connected to AC power supply.

If the fans still do not work after following the above steps, then refer to *RF Amplifier Replacement Procedures* to replace fans.

4- FUSE AND FUSE HOLDER FUNCTIONAL CHECK

1. Visually check that fans are rotating.

2. If fans are not rotating, check the following:

   a) Check fuse holder – make sure fuse holder is inserted in back of RF Amplifier.
   b) Check fuses – make sure fuses are not broken.

If the fuses and/or fuse holder needs to be replaced, refer to RF Amplifier Replacement Procedure to replace fuses or fuse holder.

5- 24 VOLT POWER SUPPLY FUNCTIONAL CHECK

1. Check the LED light panel on front of RF Amplifier. The following may indicate the 24 Volt power supply is not functioning properly:

   - POWER: Light OFF when amplifier is powered indicates that the 24Volt Power Supply is not functioning properly.
   - VDD: Light OFF when amplifier is powered indicates that the 24Volt Power Supply is not functioning properly.
   - READY: Light OFF when amplifier is powered may indicate that the 24Volt Power Supply is not functioning properly.

If the above lights are OFF, then refer to *RF Amplifier Replacement Procedures* to replace 24 Volt Power Supply.

6- 55 VOLT POWER SUPPLY FUNCTIONAL CHECK

1. Check the LED light panel on front of RF Amplifier. The following may indicate the 55 Volt power supply is not functioning properly:
    - VDD: Light OFF when amplifier is powered indicates that the 55Volt Power Supply is not functioning properly.
    - READY: Light OFF when amplifier is powered may indicate that the 55Volt Power Supply is not functioning properly.

If the above lights are OFF, then refer to *RF Amplifier Replacement Procedures* to replace 55 Volt Power Supply.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	May 13, 1998	K. Keshena	Preliminary version.
0	June 30, 1998	K. Keshena	Initial release.
1	August 5, 1998	K. Keshena	Updated scan protocol.